System Architecture & Deployment Blueprint

Sovereign Digital Public Infrastructure (DPI) for African Industrial Supply Chains

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1. Executive Architectural Philosophy

Traditional enterprise software stacks depend heavily on centralized, multi-layered foreign cloud middleware. This introduces:

1. **Volatile Foreign-Currency Subscriptions:** Recurring USD/EUR licensing leaks that drain corporate transformation, SLP, and ESD budgets.
2. **Perimeter Attack Vectors:** Transnational data backhauling that exposes sensitive vendor data to offshore interception and cloud subpoenas.
3. **Regulatory Friction:** Substantial non-compliance risk under South Africa's **Protection of Personal Information Act (POPIA Act 4 of 2013)**.

The NSK Sovereign Paradigm

NSK Devpreneur Hub implements an **on-premise, middleware-free, local-first system architecture**. We translate complex enterprise and municipal validation workflows into ultra-lightweight execution layers that process high-throughput data securely within resource-constrained, physical operational environments.

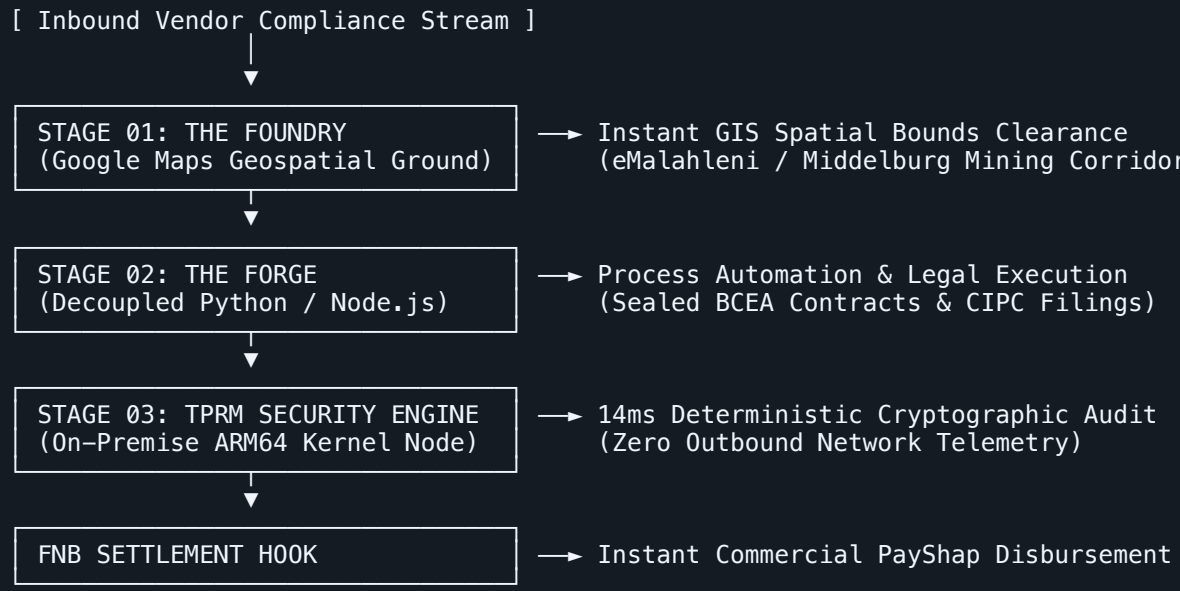
2. Physical Edge Hardware Deployment Layer

The production environment bypasses international cloud dependencies by deploying directly on an on-premise, physical edge computing cluster:

- **Target Infrastructure:** Localized **3x Orange Pi 5 Plus Single-Board Computer (SBC)** array.
- **Hardware Profile (Per Node):** Rockchip RK3588 octa-core processor (4x Cortex-A76 @ 2.4GHz + 4x Cortex-A55 @ 1.8GHz), integrated 6 TOPS NPU, and 16GB LPDDR4X RAM partition.
- **Cluster Operational Roles:** The 3 physical units operate as an isolated, high-efficiency sandboxing and transaction-processing environment. They execute cryptographic workflows directly inside physical mine administrative perimeters or municipal data hubs.
- **Resilience & Power Footprint:** The entire cluster draws **under 15W of total power**, operable continuously on local solar and battery backup through Stage 6 load shedding.

3. The Decoupled Continuous Triad

The backend is architected in decoupled native Python and Node.js microservices that communicate via lightweight internal mechanisms with **zero external API dependencies**:



4. Enterprise Firewall & Corporate Network Safety

- **Zero-Inbound Port Exposure:** LaunchSense and the TPRM engine do not require open inbound ports or external incoming traffic tunnels to function.
- **Strict Data Isolation:** All database modifications, vendor verification routines, and CIPC administrative pipeline models are contained entirely inside the local node network.
- **Air-Gapped Telemetry:** Zero bytes of vendor PII or corporate metrics leak across international borders or outside the client's internal LAN.

Live Production Verification:
• Interactive Console: `https://devpreneur-hub.vercel.app/tprm-audit`
• Enterprise Portal: `https://devpreneur-hub.vercel.app/enterprise`
• Code Repository: `https://github.com/neferititiSwaq/nsk-tprm-security-engine`